Renwei Meng

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EDUCATION

Anhui University, Hefei, China — Sep. 2023 – Jun. 2027
B.Eng. in Digital Media Technology — **GPA: 3.95/4.0**; **Rank: 1/100**; **IELTS: 7.0**
Teaching Assistant: Data Structures.
Relevant Coursework: Machine Learning, Probability and Statistics, Linear Algebra, Calculus I–III, Discrete Mathematics, Data Structures, Data Analysis, Operating Systems.

RESEARCH INTERESTS

Reinforcement learning and post-training for foundation models and agents, focusing on exploration, credit assignment, and reliable learning from feedback.
Multimodal reasoning and world models, particularly using auxiliary supervision to improve reasoning, prediction, and generalization.

PUBLICATIONS

Peer-Reviewed Conference Publications

- **FCR-PO: Learning Where to Retry in Language-Model Reasoning**. *First Author. ACML 2026.* **Accepted**.
- **Learning Which Evidence-Seeking Decisions Matter: Calibrated Counterfactual Replay for Language Agents** (PCCC-RL). *First Author. ACML 2026.* **Accepted**.
- **Know2Guess: A Contamination-Aware Multi-Zone Benchmark for Knowledge-Boundary Evaluation in Large Language Models**. *Co-first Author. ICONIP 2026.* **Accepted**.
- **Conditional Riemannian Adaptive Flow Transport for Cross-Subject EEG Decoding**. *First Author. PRICAI 2026.* **Accepted**.
- **Group Resonance Network: Learnable Prototypes and Multi-subject Resonance for EEG Emotion Recognition**. *First Author. ICANN 2026.* **Published**.
- **S-AODP++: Structure-Aware AOI-Guided Scanpath Attention for AI-Generated Image Identification**. *Co-first Author. ICMI 2026.* **Published**.
- **AuRA: Aligning Voice and Physiology by Autonomic Self-Supervision for Label-Efficient Affective and Clinical Sensing**. *First Author. BIBM 2026.* **Accepted**.
- **CausalG2C: An Intervention-Aware Genome-to-Cell Model for Zero-Shot Genetic Perturbation Prediction**. *First Author. BIBM 2026.* **Accepted**.

Manuscripts Under Review

- **Normal Fréchet Drift: Measuring Post-Training Departure from Pretrained Reference Manifold**. *Co-first Author. ICLR 2027.* **Under Review**.

Preprints

- **VulnAgent-R2: Evidence-Calibrated Multi-Agent Auditing for Repository-Level Vulnerability Detection**. *Co-first Author.* **arXiv Preprint**.
- **Explainable Innovation Engine: Dual-Tree Agent-RAG with Methods-as-Nodes and Verifiable Write-Back**. *First Author.* **arXiv Preprint**.

RESEARCH EXPERIENCE

Geometric Diagnostics for Generative Model Post-Training — Jul. 2026 – Oct. 2026
Lead Researcher | Advisor: Prof. Tao Lin | Westlake University

- Proposed **Normal Fréchet Drift (NFD)** to measure post-training departures from a pretrained reference manifold, distinguishing them from redistribution along the same manifold.
- Developed a sample-based geometric estimator with theoretical error bounds, separating reconstruction bias from sampling uncertainty.
- Found the largest drift in FlowDPO among evaluated SD3.5-M checkpoints; lower-NFD images received **84.4%** of overall-quality preferences in 120 selected blinded pairs.

Reliable Reinforcement Learning for Language Models and Agents — Jan. 2026 – Jul. 2026
Independent Research

- Developed **FCR-PO** to learn which reasoning steps are worth retrying by comparing targeted intervention with ordinary continuation.
- Achieved **77.70%** mean accuracy versus **71.03% for BranPO-adapt** on MGSM, ProofWriter, and Spider at the same 300M-token budget; retained a **5.44-point** gain over random retry selection with matched record counts.
- Developed **PCCC-RL** to identify which evidence-seeking decisions yield correct, fully supported answers through counterfactual replay.
- Improved joint answer–evidence exact match by **4.19 / 3.86 points** on HotpotQA / MuSiQue over an access-matched replay baseline at the same 100M-token budget.

RESEARCH EXPERIENCE (CONTINUED)

Knowledge-Boundary Evaluation for Language Models Jun. – Sep. 2025
Researcher | Anhui University

- Developed **Know2Guess**, a 1,200-item benchmark of when language models should answer or abstain, with analyses of prompt sensitivity, contamination risk, and benchmark construction artifacts.

Brain-Guided Post-Training for Visual Reasoning and World Models Oct. 2026 – Present
Lead Researcher | Advisor: Zhi Lu | Tsinghua University

- Developing methods that use human EEG and fMRI as additional post-training signals for visual reasoning and physical prediction, without requiring brain recordings at inference.
- Developed **RELAY** to transfer temporal EEG structure into visual reasoning; improved held-out pattern-family accuracy from **60.35% to 65.97%**, with a **2.90-point** gain on Qwen2.5-VL-3B.
- Developed **SCOPE** to prioritize world-model prediction errors using brain responses; improved Physion contact accuracy from **70.58% to 78.62%** and PushT success from **62.50% to 70.40%**.

Multimodal Post-Training and AI for Science Sep. 2025 – Jul. 2026
Lead Researcher | Advisor: Prof. Shiang Hu | Anhui University

- Developed **AuRA** and co-developed **S-AODP++**, using physiological and human-gaze signals as additional training supervision with pretrained speech and vision models, without requiring these signals at inference.
- Also developed **CausalG2C** for genetic perturbation prediction, **CRAFT** for cross-subject EEG decoding, and **GRN** for EEG emotion recognition.

Evidence-Calibrated Agents for Software Vulnerability Detection Mar. 2025 – Jan. 2026
Lead Researcher | Advisor: Chen Zhang | Anhui University
National College Student Innovation Program

- Developed **VulnAgent-R2** for repository-level vulnerability auditing with evidence checking and selective verification; achieved **0.606 F1** on JITVul with **38.3%** fewer online tokens than always-full multi-agent execution.

HONORS & AWARDS

- **National Scholarship** 2025
- **Finalist**, Mathematical Contest in Modeling (MCM/ICM) 2026
- **National Second Prize**, National College Students Market Survey and Analysis Competition 2026
- **First Prize**, National College Student Mathematical Modeling Competition (Provincial) 2025

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Java, R, SQL, Bash / zsh.
Machine Learning & Agents: PyTorch, TensorFlow, LangChain, LangGraph.
Research Infrastructure: Linux, Git, Docker, LaTeX.

ADDITIONAL EXPERIENCE

- **Java Backend Developer Intern**, Taiyuan Longwei Electronic Technology Co., Ltd. Jul. – Sep. 2025
  Built Spring Boot / MyBatis services and RESTful APIs; coordinated database and frontend integration.
- **Systems Operations Intern**, State Grid Shanxi Electric Power Co., Ltd. Jul. – Sep. 2024
  Maintained power-system services and analyzed operational metrics for reliability reports.
- **Coronary CTA Artifact Removal** — Lead Researcher, Anhui University. Jan. – Jun. 2025
  Advisor: Prof. Zhe Jin
  Combined U-Net and GAN denoising to improve SNR and SSIM over conventional filters.
- **Carbon Price Prediction** — Lead Researcher, Anhui University. Jun. 2024 – Jun. 2025
  Advisor: Prof. Zhang-HangJian Chen
  Developed a CNN–recurrent attention model across seven Chinese carbon markets.

PATENTS

- **A generative deepfake video detection method and its system**. Co-inventor.
  Chinese invention patent application No. 202610165744.1.
- **Resting-state EEG quality assessment based on dual-branch contrastive learning**. Co-inventor.
  Chinese invention patent application No. 202512044739.3.